

Connecting Flask Application with MySQL Database

(Simple Notes in English)

What is the Purpose?

We use MySQL database to store application data permanently.

Example:

- Student details
- Login information
- Product details
- Orders
- Employee records

A Flask application communicates with MySQL using a connector/driver.

Technologies Used

Technology	Purpose
Flask	Backend Web Framework
MySQL	Database
mysql-connector-python	Connector/Driver between Flask and MySQL
HTTP	Communication between Browser and Flask App

Architecture Flow

Browser
↓ HTTP Request
Flask Application
↓
MySQL Connector
↓
MySQL Database

Step 1: Install MySQL Connector

Open terminal and run:

```
pip install mysql-connector-python
```

Purpose

This package helps Python/Flask communicate with MySQL database.

Step 2: Import MySQL Connector

```
import mysql.connector
```

Purpose

Imports MySQL connector library into Flask application.

Step 3: Establish Connection with MySQL Database

```
con = mysql.connector.connect(  
    host="localhost",  
    user="root",  
    password="Jani@249",  
    database="college"  
)
```

Explanation

Parameter Meaning

host	MySQL server location
user	MySQL username
password	MySQL password
database	Database name

Step 4: Create Cursor Object

```
cursor = con.cursor()
```

Purpose of Cursor

Cursor is used to:

- Send SQL queries from Flask to MySQL
 - Execute queries
 - Fetch results
-

Step 5: Execute SQL Queries

Insert Data

```
cursor.execute(  
    "INSERT INTO students(id, name) VALUES(1, 'Jani')"  
)  
con.commit()
```

Read Data (SELECT)

```
cursor.execute("SELECT * FROM students")
```

Update Data

```
cursor.execute(  
    "UPDATE students SET name='Basha' WHERE id=1"  
)  
con.commit()
```

Delete Data

```
cursor.execute(  
    "DELETE FROM students WHERE id=1"  
)  
con.commit()
```

Step 6: Fetch Result from Database

Fetch All Records

```
result = cursor.fetchall()
```

```
print(result)
```

Example Output

```
[(1, 'Jani'), (2, 'Rahul')]
```

Fetch One Record

```
result = cursor.fetchone()
```

```
print(result)
```

Example Output

```
(1, 'Jani')
```

Step 7: Close Cursor and Connection

```
cursor.close()
```

```
con.close()
```

Purpose

Function	Purpose
----------	---------

cursor.close()	Closes cursor object
----------------	----------------------

con.close()	Closes database connection
-------------	----------------------------

Closing connection helps:

- Save memory
 - Improve performance
 - Avoid database issues
-

CRUD Operations

Operation	SQL Command	Meaning
-----------	-------------	---------

C	CREATE / INSERT	Add data
---	-----------------	----------

R	READ / SELECT	Read data
---	---------------	-----------

U	UPDATE	Modify data
---	--------	-------------

D	DELETE	Remove data
---	--------	-------------

Complete Example

```
import mysql.connector

# Database Connection
con = mysql.connector.connect(
    host="localhost",
    user="root",
    password="Jani@249",
    database="college"
)

# Create Cursor
cursor = con.cursor()

# Execute Query
cursor.execute("SELECT * FROM students")

# Fetch Data
result = cursor.fetchall()

# Print Result
print(result)

# Close Connection
cursor.close()
con.close()
```

Real-Time Flow

User → Browser → HTTP Request →
Flask App → MySQL Connector →
MySQL Database

Database Response →
Flask App →
Browser →
User

Important Interview Points

Why do we use MySQL Connector?

To connect Python/Flask application with MySQL database.

What is Cursor?

Cursor is an object used to execute SQL queries and fetch results.

Why use commit()?

To permanently save INSERT, UPDATE, DELETE changes in database.

Difference Between fetchone() and fetchall()

Method	Purpose
---------------	----------------

fetchone()	Gets single row
------------	-----------------

fetchall()	Gets all rows
------------	---------------

Simple Summary

1. Install connector
2. Import connector
3. Create database connection
4. Create cursor
5. Execute queries
6. Fetch results
7. Close connection